

Wall Vorticity Boundary Conditions in Streamfunction-Vorticity Formulation

1 General Derivation

For incompressible flow,

$$\nabla^2 \psi = \omega \quad (1)$$

Expand the first interior point about a wall point:

$$\psi_p = \psi_w + \left(\frac{\partial \psi}{\partial n} \right)_w h + \left(\frac{\partial^2 \psi}{\partial n^2} \right)_w \frac{h^2}{2} + O(h^3) \quad (2)$$

Using

$$\left(\frac{\partial \psi}{\partial n} \right)_w = U_t \quad (3)$$

and

$$\left(\frac{\partial^2 \psi}{\partial n^2} \right)_w = \omega_w \quad (4)$$

we obtain

$$\psi_p = \psi_w + U_t h + \omega_w \frac{h^2}{2} \quad (5)$$

Hence

$$\boxed{\omega_w = \frac{2(\psi_p - \psi_w)}{h^2} - \frac{2U_t}{h}} \quad (6)$$

2 Individual Walls

Bottom wall:

$$\omega_{i,0} = \frac{2\psi_{i,1}}{h^2}$$

Top moving lid:

$$\omega_{i,ny-1} = \frac{2\psi_{i,ny-2}}{h^2} + \frac{2U_{lid}}{h}$$

Left wall:

$$\omega_{0,j} = \frac{2\psi_{1,j}}{h^2}$$

Right wall:

$$\omega_{nx-1,j} = \frac{2\psi_{nx-2,j}}{h^2}$$

3 Summary

Wall	Vorticity BC
Bottom	$\omega_{i,0} = 2\psi_{i,1}/h^2$
Top	$\omega_{i,ny-1} = 2\psi_{i,ny-2}/h^2 + 2U_{lid}/h$
Left	$\omega_{0,j} = 2\psi_{1,j}/h^2$
Right	$\omega_{nx-1,j} = 2\psi_{nx-2,j}/h^2$